

Shielding QR Codes: Unveiling the Real-World Illicit Promotion Behind Adversarial QR Codes

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Abstract

Adversarial QR Code Images (AQRI) represent an emerging threat vector for covert (usually illicit) online promotion. They use adversarial perturbations to evade QR detectors (e.g., OCR-based models) while retaining decodability for information delivery, facilitating malicious content dissemination, and posing risks to both platforms and users. Though adversarial attacks are well-studied, targeted techniques against structured QR codes are underexplored.

To systematically investigate the real-world AQRI abuse, we cooperated with a leading Internet service provider. With the help of our partner, grounded in empirical observations, we introduce Adato, an enhanced framework for AQRI detection, by prioritizing finder pattern regions and identifying adversarial techniques through cross-platform consistency checks. Experimental results demonstrate that Adato achieves 98.6% precision and 97.8% recall on AQRI detection, significantly outperforming existing detectors. With the collaboration of our partner, we legally obtained posts with images from five well-known international social media platforms from September, 2024, to March, 2025, e.g., Reddit, Baidu Tieba. We applied Adato to over 40 million images and identified 68,467 AQRI, demonstrating their widespread real-world use and their ability to evade existing moderation mechanisms. Analysis of our detected AQRI reveals that AQRI are widely used for illicit promotion: 95.78% are linked to 2,079 malicious URLs, spanning 7 business categories. Additionally, we analyzed the information dissemination strategies employed, such as redirect chains and indirect propagation paths that exploit cross-platform inconsistencies. These results highlight Adato’s effectiveness in strengthening existing moderation and recognition pipelines against AQRI abuses.¹

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¹The full version of this paper, including appendix, is available at <https://github.com/S5deded/usenix-security26-qr-codes-full-paper/blob/main/full-paper.pdf>.

1 Introduction

QR codes are a popular medium for fast and convenient information dissemination, capable of encoding various types of payload, e.g., text and URLs. Commonly, QR codes are used to direct users to external websites on social media platforms, making them an efficient cross-platform redirection mechanism. However, their encoding and redirection capabilities also make QR codes a dissemination vector for malicious content [63, 117], such as phishing links [21, 28, 65, 85, 124]. To protect users from malicious content via QR codes, major social platforms implement QR code moderation and posting restrictions, including TikTok [96], Baidu Tieba [10], and Sina Weibo [88]. Specifically, QR code moderation typically involves detecting a QR region in the image, decoding the embedded payload, and then inspecting the decoded content against security and policy rules. *Despite advancements in moderation technologies, abuse of QR codes to spread malicious content has evolved in parallel.*

Adversarial QR Code Image (AQRI). AQRI represent a new class of threats: QR codes that are visually perturbed to evade platform moderation systems while remaining decodable to end users (see Figure 3). By disrupting image-layer detection, AQRI can bypass platform-level moderation and be successfully published on social media platforms, thereby expanding the reach of illicit or policy-violating content. Unlike traditional adversarial examples that induce model misclassification through imperceptible noise, AQRI apply perceptible yet controlled transformations that preserve decodability. This constraint makes AQRI generation more targeted and practically impactful. Recent advances in generative models further exacerbate this threat by enabling AQRI to blend into complex visual backgrounds, making detection harder and evasion more effective [31, 91, 108, 113]. AQRI thus exemplify the exploitation of adversarial AI techniques in real-world malicious content dissemination campaigns.

While most social platforms have QR code moderation, they have not recognized the novel AQRI threat. Specifically, testing 4 mainstream QR detection tools and 5 platforms’

moderators on 100 in-the-wild AQRI showed poor detection rates, exposing their inability to handle this novel threat, since AI models relying on structural pixel features fail to capture obfuscated AQRI traits. It also makes conventional adversarial defense (e.g., feature alignment) ineffective [3, 14, 30, 105, 116]. Thus, *efficient, automated AQRI detection and impact assessment are urgently needed.*

To address the gap in understanding and mitigating such novel AQRI threats, we collaborated with a leading global internet service provider, Baidu². Social media’s broad public reach reflects the impact of AQRI on ordinary internet users and poses severe challenges to platform moderation. To assist moderation systems in detecting AQRI and analyzing their underlying techniques and payloads, we scope our measurement study to five high-impact, large-user-base international platforms. Specifically, our work focuses on: *How do AQRI affect existing QR code detection and moderation capabilities? What is the current usage and abuse status of AQRI in real-world scenarios? Which adversarial techniques and strategies are employed in crafting AQRI?*

Our Work: Detecting AQRI with Adato. Addressing the challenges posed by AQRI is nontrivial due to their stealthy design, black-box nature, and the limited effectiveness of existing detection methods. Motivated by empirical analysis of real-world AQRI collected from user complaints with our partner, we develop Adato (AQRI Detection and Analysis Tool), a unified framework for detecting and restoring AQRI to enable robust decoding, including 4 steps. First, Adato collects posts from 5 popular social media platforms (Baidu Tieba, Baijiahao, Reddit, X, and Instagram) via legal APIs and scraping. Second, it improves QR code detection by localizing and verifying finder pattern regions, which AQRI typically preserve for decodability. Third, Adato identifies adversarial techniques by exploiting inconsistencies across multiple QR scanners and moderation engines, and attributes each AQRI to specific manipulations techniques using statistical profiling of basic visual features. Finally, for visual restoration and decoding, Adato performs technique-aware feature-level restoration to correct perturbations and improve decoding success. Evaluated on a real-world dataset, Adato achieves 98.6% precision and 97.8% recall on AQRI detection, and reaches a 99.87% decoding success rate. Moreover, Adato has a mere 46ms processing latency for QR code detection. It meets platform requirements for large-scale moderation, improving the performance of existing detectors by an average of 5.48x while enhancing end-user decoding success rates.

Analyzing AQRI in the wild. To answer research questions for AQRI, we legally crawled 40,147,738 images from 5 social media platforms over 181 days (2024.09.13-2025.03.13), with the help of our partner. Finally, Adato identified 1,585,706 QR code images (3.94%), among which 68,467 were verified as AQRI (4.32%). It reveals that AQRI has

been widely adopted across social platforms for promotion. Furthermore, we conducted an in-depth analysis of AQRI’s manipulation techniques, embedded information, abuse patterns, and impacts. Specifically, Adato identified 13 adversarial techniques in real-world AQRI that, unlike traditional attacks requiring complex optimization [67, 68, 84, 109], rely on simple image processing such as structural distortion (e.g., geometric deformation, downscaling) and visual concealment (e.g., transparency, black masks). Though simple, their combination and fusion enable highly effective, low-cost evasion. Moreover, our analysis reveals the emerging use of generative methods, predominantly Stable Diffusion [82], in fabricating AQRI. These tools effectively blend multiple adversarial effects while preserving decodability, posing an escalating challenge to current moderation systems. Compared to non-AI counterparts, AI-generated AQRI exhibit significantly higher bypass rates (by 20.0%) and decoding success (by 18.0%) on average. The absence of usage restrictions and the accessibility of user-friendly interfaces further increase their abuse potential. In contrast, advanced Large Vision Models (LVMs) display functional blindness: although they achieve moderate bypass rates (73.5%), their end-to-end generation fails to preserve QR syntax, resulting in low decodability (13.3%) and rendering them currently impractical for adversarial use.

We further investigated the decoded information to confirm the actual content being propagated by AQRI. Our analysis revealed that AQRI were widely abused for various underground activities. Website links (99.80%) are the most common form of content dissemination. AQRI-related websites adopt diverse construction forms, from personally built sites (46.54%), abuse of short-link services (23.46%), to compromised authoritative websites (11.00%). They cover 7 malicious types, most of which are online pornography. Additionally, we found that adversaries mask malicious content with legitimate images and mimic normal posting behaviors. Furthermore, we observed a unique cross-platform strategy: leveraging varying moderation capabilities across platforms to propagate AQRI with differing adversarial intensities.

Contributions. This paper makes the following contributions:

- **Enhanced AQRI Detection and Restoration:** We propose Adato, the first dedicated tool for detecting and restoring Adversarial QR Code Images. Leveraging finder-pattern-guided geometric validation, Adato achieves robust AQRI detection with a 97.8% success rate. Adato’s restoration capability makes it a reliable pre-adaptor, boosting detection by 5.48x across five existing QR detectors, including our partner’s commercial QR processing system deployed in real-world settings. To support broader adoption, we will release Adato as an open-source tool for developers and practitioners.
- **Novel Insights of AQRI Impact:** Based on Adato, we conducted the first large-scale, systematic measurement study of real-world AQRI abuse and constructed the first annotated AQRI dataset. Our analysis provides new insights into AQRI, revealing prevalent evasion strategies, common manipulation

²Our collaborator provides multiple internet services, including famous social media platforms.

techniques, and their tangible impacts on mainstream QR processing platforms. We plan to share the dataset with researchers upon request to catalyze future research in AQRI analysis and defense.

2 Background

In this section, we first introduce QR codes and how they are disseminated and moderated on online platforms, then define adversarial QR code images (AQRI), present our threat model, and clarify the scope of our study.

2.1 QR Code: Concept and Properties

QR codes are matrix barcodes with information-carrying pixel distributions, available in versions 1-40 [98]. Higher versions use more modules for larger capacity, increasing graphical complexity. Despite varying content-dependent pixel layouts, standardized design specifications require two key components: finder pattern and encoding region [98]. A QR code typically includes three finder patterns (top-left, top-right, and bottom-left), which determine its boundary and orientation. The encoding region, enclosed by these finder patterns, encodes information through a prescribed arrangement of pixels.

A functional QR code must satisfy two properties: *detectability* and *decodability*, which correspond to the two stages of QR processing. **Detectability** means a QR tool can reliably determine *whether a QR code is present and localize its region* (e.g., estimate its bounding quadrilateral and orientation). **Decodability** means the tool can correctly *recover the embedded payload*. Both properties rely on preserving key structural constraints: the finder patterns must remain sufficiently intact for localization. While detectability and decodability are fundamental properties of QR codes, their realization in practice depends on the specific detection and decoding systems in use. In standard QR processing, detection and decoding are *sequential* operations: a QR code must first be *detected* before its payload can be *decoded*. After successful detection and decoding, a QR code can reveal payloads such as URLs, plain text, or base64-encoded data, making it a widely used medium for promotion and information dissemination.

2.2 QR Code Dissemination and Moderation

QR codes are widely shared on many platforms, with social media being a primary vector (e.g., TikTok [96], Instagram [45], and Sina Weibo [88]). To mitigate malicious dissemination, external redirection, and unauthorized advertising, many platforms impose explicit or implicit restrictions on QR-code usage. By analyzing the Terms of Use (ToU) of eight major platforms, we identify three regulatory stances: (1) Baidu Tieba, Sina Weibo, Zhihu, and Xiaohongshu explicitly prohibit QR code postings to prevent external redi-

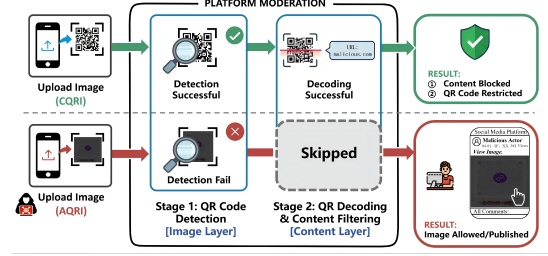


Figure 1: Overview of the Platform Moderation Pipeline and AQRI Evasion Mechanism.

rection [11, 89, 112, 123], making the mere presence of a QR code a policy violation, regardless of its payload; (2) TikTok bans QR codes that embed malicious or prohibited content (e.g., malware, pornography, gambling, phishing, or unauthorized commercial links) [97], and external reports suggest that videos with QR codes may be excluded from recommendations [69]; (3) Instagram, X (formerly Twitter), and Reddit do not state explicit policies, but our measurements indicate implicit restrictions. For instance, Instagram limits scanning to proprietary in-app codes [66], and some subreddits ban external links, including short URLs and QR codes [81]. Beyond platform-specific policies, QR detection is also supported by general-purpose moderation services such as Google Cloud Vision AI [36] and Baidu AI Cloud [8].

Moderation Pipeline. As shown in Figure 1, the moderation process typically involves two steps.

- **Step 1: QR Code Detection (Image Layer):** Platforms first detect the presence and location of QR codes within images using object detection techniques. Early rule-based methods (e.g., fixed pixel distributions, square edges, ISO patterns) [12, 24, 86, 90] have largely been replaced by deep learning models such as R-CNN [32, 33] and Transformers [103], which offer higher accuracy. Detected regions may undergo image enhancement (e.g., binarization, geometric correction, grid alignment) to support reliable decoding [18, 40, 56].
- **Step 2: QR Decoding & Content Filtering (Content Layer):** Once the QR region is localized and preprocessed, the platform extracts the embedded payload using a QR decoder. This content typically includes URLs or plain text. Common techniques include blacklist matching and machine learning classifiers to detect malicious links, phishing attempts, or unauthorized promotions [50, 53, 83].

Notably, platforms implement these moderation steps differently according to their policies. Platforms that ban all QR codes may rely only on Step 1 (detection), while those targeting malicious payloads use both Step 1 and Step 2 (decoding and content filtering). Despite these variations, the underlying moderation pipeline follows the same sequential flow.

2.3 AQRI: Definition and Threat Model

Adversarial QR Code Images (AQRI) are crafted to evade platform QR-code moderation for covert or broader dissemination reach, while remaining decodable by end users to ensure successful information delivery. Unlike traditional adversarial examples that use imperceptible perturbations to induce misclassification [34, 54, 72, 73, 92], AQRI apply low-complexity, coarse-grained transformations (e.g., masking, distortion, occlusion) that exploit weaknesses in automated detection systems, while remaining decodable by end users regardless of human perception and visual usability.

Threat Model. We formalize the threat model of AQRI to precisely characterize their adversarial objective. Let $x \in \mathcal{X}$ denote a conventional QR code image (CQRI), encoding a message $m \in \mathcal{M}$. Let $\text{Detect}_S : \mathcal{X} \rightarrow \{0, 1\}$ be the detection function employed by a QR processing system S (e.g., a platform or a QR processing tool), where $\text{Detect}_S(x) = 1$ indicates that a QR code is successfully localized in the image. Once detected, the system applies a decoding function $\text{Decode}_S : \mathcal{X} \rightarrow \mathcal{M} \cup \{\perp\}$ to extract the embedded message, with $\perp$ indicating decoding failure. Thus, we expect $\text{Detect}_S(x) = 1$ and $\text{Decode}_S(x) = m$ for a valid CQRI. Let $\delta : \mathcal{X} \rightarrow \mathcal{X}$ be a transformation function used by the adversary to generate an AQRI $x^{\text{adv}} = \delta(x)$, which is then published on platform P and satisfies the following conditions:

$$\text{Detect}_P(x^{\text{adv}}) = 0, \text{Detect}_Q(x^{\text{adv}}) = 1, \text{Decode}_Q(x^{\text{adv}}) = m \quad (1)$$

for some tool $Q \in \mathcal{D}$, where $\mathcal{D}$ denotes the set of potential QR processing systems, including external or end-user applications. This setting reflects the **transmission model**, where the AQRI is *undetectable* on the publishing platform P , thereby skipping the decoding stage, yet remains *detectable* and *decodable* by external tools—enabling it to bypass moderation while still delivering its payload (usually illicit) to end users. This transmission model stems from differences in the capabilities of QR code processing tools.

- **Adversarial roles and objectives.** In this model, the adversary is the party who generates and publishes AQRI, typically pursuing two objectives: 1) bypassing image-layer moderation (Step 1) to publish the QR code and expand its dissemination scope; and 2) ensuring the embedded message remains decodable by end users, thereby completing the information delivery process despite evading platform detection. The victims include both the platform, whose moderation pipeline is circumvented, and the users, who may unknowingly receive unreviewed or policy-violating content.

This threat model reflects real-world abuse scenarios in which adversaries exploit inconsistencies across platform-specific detection and decoding systems. Our preliminary measurement study (see Section 3) highlights this gap, showing how a QR code can evade moderation on the host platform but remains functional for end-user—thereby enabling covert or policy-violating dissemination.

Research Scope. This study investigates AQRI as a distinct threat vector that targets the image-layer detection stage of

platform moderation pipelines. While prior work has largely focused on bypassing content-layer checks through payload obfuscation (e.g., concealing malicious URLs), our work highlights a more fundamental evasion strategy: perturbing the visual structure of QR codes to evade initial detection altogether. Rather than introducing new attacks, our study analyzes AQRI already used in the wild. Specifically, through formal collaboration with a major global network service provider, we curated a high-quality ground-truth dataset of user-reported, manually verified AQRI and analyzed their impact on moderation pipelines (Section 3). We then characterize the adversarial properties of these AQRI by comparing them to CQRI, using a set of interpretable visual metrics to quantify key structural and perceptual differences and support threat landscape analysis (Section 4).

3 Preliminary Study of AQRI

To empirically validate the threat model of AQRI, we conduct a comprehensive measurement study using a real-world dataset. As AQRI are inherently difficult to collect from public sources due to their ability to evade standard QR detection, we collaborated with a major global network service provider to curate user-reported problematic QR codes via their complaint-handling infrastructure. We then examine the impact of AQRI on three key components of QR code moderation pipelines: detection capability, decoding reliability, and overall moderation effectiveness.

AQRI Ground-truth Dataset Construction. With the assistance of our partner, we collected 5,000 QR code-related images from real-world user complaints spanning August to September 2024. To ensure data quality, these samples were initially filtered via the platform’s predefined spam and malicious QR tags before undergoing our manual review. To ensure image quality, we first filtered out images with widths or heights below 300 pixels. Further, two expert researchers independently verified the adversarial characteristics: images exhibiting signs of manipulation through image processing techniques and a visibly disrupted or altered appearance were labeled as AQRI. Any disagreements were resolved through arbitration by a third expert. As a result, we construct a ground-truth dataset of 3,000 real-world AQRI.

To evaluate their effectiveness in bypassing moderation pipelines, we test detection performance across 4 commercial QR detectors, examine decodability using 10 widely-used decoding tools, and assess moderation responses by actively posting AQRI to 5 major social media platforms. Due to platform usage limitations, we randomly selected 100 AQRI from our dataset and paired them with 100 CQRI sampled from a public dataset [20] for comparative analysis.

AQRI Impact on Detection Capability. We assess the adversarial impact of AQRI on detection capability using 4 mainstream QR code detection services (Table 1). Specifically, we submitted each image to the corresponding service

Table 1: Detecting Success Rate of CQRI and AQRI on public QR code detection service.

Tool	Detecting Success Rate (%)	
	CQRI	AQRI
Google Vision AI [36]	95.0	11.0
Baidu AI Cloud [8]	93.0	10.0
Tencent Cloud [95]	96.0	13.0
Aliyun [4]	96.0	10.0

* : The QR code detector is powered by a large vision model.

Table 2: Decoding success rates of various decoders on AQRI.

App-Decoder	Rate (%)	Library	Rate (%)
WeChat [93]	86.0	Qreader [26]	56.0
Alipay [5]	75.0	OpenCV-WeChat [71]	55.0
QQ [94]	48.0	Zbar [48]	41.0
iOS Scanner	67.0	Zxing [118]	38.0
Google Chrome [35]	38.0	BoofCV [76]	33.0

interface and recorded the outputs, which varied in format across systems: for instance, Baidu AI Cloud returns a confidence score, whereas Google Vision AI provides tiered classification labels (e.g., UNKNOWN to VERY LIKELY). Following the criteria used in prior work [38], we treat a QR code as successfully detected if the confidence score exceeds 0.5 or the classification label is “POSSIBLE” or “VERY LIKELY”.

Table 1 shows that QR code detectors perform well on CQRIs but struggle significantly with AQRI. All services detect no more than 13% of AQRI, while achieving over 93% on CQRIs. These results indicate that current detection systems are ill-equipped to handle AQRI and fall short of the reliability for real-world QR code detection.

AQRI’s Decodeability. We conducted experiments using 10 widely adopted QR code processing tools, including 5 end-user scanning applications and 5 open-source QR decoding libraries (Table 2). While end-user applications support both detection and decoding, detection is typically user-assisted, requiring manual alignment of the QR code within a predefined scanning frame. An AQRI is defined as decodable by a tool if its content is retrieved within 3 seconds. We thus evaluate *sample-level success* (decodable by at least one tool) and *tool-level success* (the percentage of sample-level successful AQRI that a specific tool can decode, as reported in Table 2).

As shown in Table 2, although AQRI are designed to evade automated detection, they remain decodable in practice. Every AQRI in our sample was successfully decoded by at least one tool. WeChat, Alipay, and the iOS Scanner achieved the highest success rates and accounted for most decoding successes, reflecting their widespread use and robust performance. Nonetheless, manual inspection revealed that these tools struggle with specific AQRI variants, such as those that combine overexposure with density compression, highlighting their limited resilience to diverse adversarial transformations.

Table 3: Posting Success Rate of CQRI and AQRI on different social platforms.

Platform	Posting Success Rate (%)	
	CQRI	AQRI
TikTok	0.0	82.0
Weibo	0.0	71.0
Xiaohongshu	0.0	75.0
Baidu Tieba	0.0	84.0
Zhihu	0.0	91.0

A detailed analysis of these transformations and their effects on detection and decoding is provided in Section 5.2.

AQRI Impact on Platform Moderation. We further conduct active experiments by posting both AQRI and CQRIs to social media platforms to evaluate their direct impact on platform moderation effectiveness. We targeted five platforms known to moderate QR code content (Table 3) and recorded the success rate of each post. If the QR code is detected, the post will not be successfully published. In our active experiments, the QR code samples were constructed by mimicking real AQRI only at the image level (i.e., visual appearance). The decoded payloads were entirely benign. Specifically, each QR code contained only plain text stating the purpose of the experiment and providing our contact information, with no malicious URLs, spam content, or harmful payloads. To prevent user confusion and minimize moderation burden, all posts were removed within one minute, and the experiments were conducted under the supervision of our collaborator’s legal team (Ethical Considerations).

Table 3 summarizes the posting success rates across platforms. All CQRIs were successfully blocked on every platform, resulting in a 0% posting success rate, which is consistent with stated platform policies on QR code restrictions (see Section 2.2). In contrast, AQRI achieved a posting success rate of 80.3%, revealing a substantial gap in current moderation overall capabilities against adversarial QR codes.

Conclusion. Our experiments support the AQRI threat model by revealing the asymmetry between platform-side detection and end-user decoding. AQRI can evade image-level detection while remaining reliably decodable by common user tools, enabling the covert dissemination of policy-violating content that circumvents platform safeguards yet remains accessible to recipients. In addition, our analysis of undetected AQRI yields two key observations that may inform the development of more robust QR code detection mechanisms.

Key observations on AQRI. First, to preserve their core function as information carriers ($\text{Decode}_Q(x^{\text{adv}}) = m$), AQRI consistently retain essential structural features of QR codes, particularly the finder patterns, despite adversarial modifications. Second, their primary distinction from conventional QR images lies in their ability to evade detection; detection failure itself ($\text{Detect}_P(x^{\text{adv}}) = 0$) may thus serve as a useful

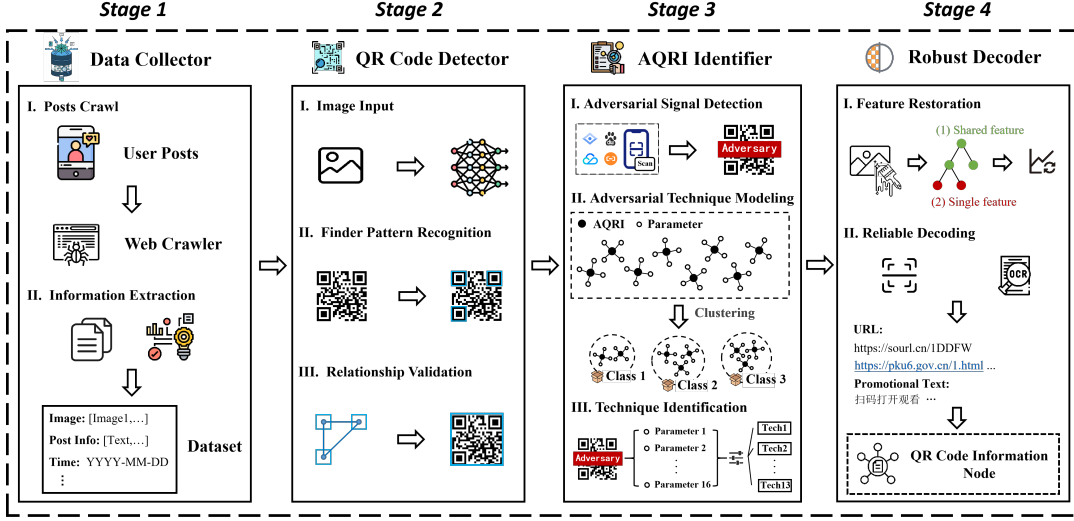


Figure 2: Workflow of Adato.

signal of adversarial manipulation.

4 Detect AQRI In The Wild

To address the growing AQRI threat, we design and develop **Adato** (**A**QRI **D**etection and **A**nalysis **T**ool (Figure 2), a unified framework for enhanced adversarial signal detection, and visual restoration to enable robust decoding. Motivated by two empirical insights from real-world AQRI (Section 3), we propose a four-stage pipeline for robust AQRI detection, characterization, and decoding: (1) collect image-based posts from popular social media platforms via APIs and scraping to build an analysis database; (2) localize QR codes using resilient geometric cues, especially corner finder patterns; (3) flag adversarial cases by cross-platform scan inconsistencies and identify the underlying manipulation techniques; and (4) apply technique-specific restoration guided by normal/adversarial feature ranges to recover CQRI-like structure, then decode with standard tools for further analysis. Importantly, Adato can function as a standalone tool or as a preprocessing module that augments existing QR scanning and content moderation systems, thereby improving their capabilities against adversarial manipulation. Below, we elaborate on each step in detail.

4.1 Data Collection

To maximize coverage, we targeted high-traffic, image-focused communities and hashtags across 5 popular social media platforms. For Baidu Tieba and Reddit, we selected the top 10 communities by traffic plus image-centric groups, totaling 35 communities (maximum traffic 17.32M visits). For Baijiahao, Instagram, and Twitter³, we combined 15 QR-

³Twitter is also known as X

related keywords from the Ground-truth dataset (Section 3) with the top 50 trending platform-specific hashtags [2, 100], yielding 65 keywords per platform. We collected posts on Reddit, Instagram, and Twitter via official APIs [46, 79, 110] using hashtag search and community crawling. As Baidu Tieba and Baijiahao provide no official API, we built a custom scraper with rate limiting (e.g., sleep intervals) for low-volume, long-horizon collection. From each post, we extracted image URLs and associated text to form the detection dataset. Regarding data analysis and storage, we used only publicly available data anonymized for technical analysis with no PII, and all procedures were supervised by our collaborator’s legal department (see Ethical Considerations).

As shown in Table 4, we constructed a large-scale dataset by crawling image-containing posts from five widely used social media platforms: Baidu Tieba, Baidu Baijiahao, Reddit, Twitter, and Instagram. The data collection process spanned 181 consecutive days, from September 13, 2024 to March 13, 2025. During this period, we harvested a total of 24,311,446 posts, from which we extracted 40,147,738 image data points. This extensive dataset captures a broad and diverse sample of user-generated visual content across multiple platforms and regions, enabling a thorough examination of AQRI prevalence, characteristics, and evasion behaviors in the wild.

4.2 QR Code Detector: Toward Robust AQRI Detection

After QR image collection, Adato determines whether an image contains a QR code and localizes its region. As shown in Figure 2, QR decoding relies on the integrity of the three corner finder patterns, which QR standards require to remain largely intact [1]. Accordingly, adversaries typically preserve

Table 4: Distribution of AQRI Detections across Platforms

Platform	# Post	# Image	# QR code	# AQRI	# Post with AQRI
Baijiahao [9]	10,872,307	16,308,460	704,511	30,566	28,563
Baidu Tieba [10]	7,083,573	11,758,731	432,023	19,388	18,730
Reddit [80]	2,618,265	4,922,338	174,054	8,342	7,056
Instagram [45]	2,117,391	4,563,912	138,191	5,713	5,600
Twitter [101]	1,619,910	2,594,297	136,927	4,458	4,334
All	24,311,446	40,147,738	1,585,706	68,467	64,283

these regions to keep AQRI decodable (Section 3). Adato exploits this invariance by locating the finder patterns and validating their geometric and spatial configuration against the standard layout. We elaborate on the two steps below.

1) Finder Pattern Region Detection: To detect finder pattern regions, we use a YOLOv8-based detector [102] customized for QR structures: we replace the standard IoU loss with Wise-IoU [99], $\text{Wise-IoU} = (1 - \text{IoU}) \cdot w$, where the dynamic weight w emphasizes small-object localization to improve coordinate precision on compact, high-contrast finder patterns; we also deepen the backbone with additional convolutional layers for finer texture features and add spatial attention to suppress adversarial noise and background clutter. The spatial attention module applies channel-wise average and max pooling, concatenates the pooled maps, passes them through a convolution layer (e.g., a 7×7 kernel), and applies a sigmoid to produce a 2D attention map. The attention map is then multiplied element-wise with the (input or intermediate) feature maps to reweight spatial locations, emphasizing finder pattern regions while suppressing irrelevant background clutter and adversarial artifacts. We train the detector for 160 epochs on our annotated dataset and observe stable convergence; at inference time, given a single image, the model outputs a structured text file specifying the detected finder pattern regions as rectangular bounding boxes.

2) Positional and Spatial Relationship Validation: After detecting finder pattern regions, the system applies a rule-based geometric verification procedure to determine whether the detections form a valid QR code layout. Following QR standards, it requires (i) at least three detected finder patterns, (ii) three non-overlapping regions of approximately equal size, and (iii) centroids of any such triple that form an approximately right-angled triangle consistent with the canonical finder-pattern geometry. If these conditions hold, the image is classified as containing a QR code, and the system estimates the full QR bounding box by geometrically aligning the verified finder patterns, using the top-left and bottom-right coordinates as location indicators. The localized QR region then supports downstream adversarial technique analysis and precise cropping for decoding, ensuring that only structurally valid QR codes advance to subsequent restoration stages.

4.3 AQRI Identifier: Technique Discovery and Identification

Adversarial Signal Detection. To determine whether a QR image has been adversarially manipulated, we propose an adversarial-signal detection method that exploits discrepancies across multiple QR detection tools. The method is motivated by a key observation: unlike CQRIs, AQRI may evade detection, and such failures can signal adversarial behavior. We instantiate this method using a real-world detector suite comprising four commercial QR detection services (Google Cloud Vision AI, Baidu AI Cloud, Tencent Cloud, and Aliyun). We selected these detectors based on their prevalence and empirical performance (Section 3), prioritizing those more susceptible to AQRI to maximize coverage of realistic evasion cases. Any sample that bypasses at least one detector is treated as an adversarial candidate.

Adversarial Technique Modeling. We present a data-driven methodology to analyze and identify adversarial techniques in AQRI based on their visual and structural distortions. In this work, an adversarial technique denotes an effect-centric distortion pattern identifiable from the final image, which we infer by clustering observed AQRI samples according to their distortion characteristics. Specifically, we construct a 16-dimensional feature space $\mathcal{V} = \mathbb{R}^{16}$ capturing visual and structural attributes commonly perturbed in AQRI. Appendix B details the features, including metrics capturing aspects of contrast, color distribution, edge clarity, finder-pattern visibility, and image region occupancy. We map each QR code image $x \in \mathcal{X}$ to a feature vector $v(x) \in \mathcal{V}$ via a feature extraction function $v: \mathcal{X} \rightarrow \mathcal{V}$. Using these representations, we apply unsupervised clustering to a dataset of real-world AQRI (Section 4.5), obtaining 23 clusters that group images with similar distortion patterns. We then manually inspect each cluster’s centroid and representative samples to identify recurring manipulation patterns, which yields 13 distinct adversarial techniques (Section 5.2). For clarity, we name each technique after the closest interpretable image operation that captures the corresponding pattern.

Feature-based Adversarial Technique Identification. Unsupervised clustering assigns each image to a single cluster, which prevents comprehensive technique attribution when an AQRI exhibits multiple adversarial techniques. To address this limitation, we propose a feature-based identification method that defines technique-specific adversarial ranges over the feature dimensions and enables multi-label attribution by comparing an image’s features against these ranges.

• *Modeling Adversarial Feature Ranges:* We model each feature’s statistical distribution to estimate both its normal range (from CQRIs) and its technique-specific adversarial ranges. For each feature $i \in [1, 16]$, we define the universe of valid values as $\mathcal{U}_i$. From a collection of CQRIs, we model the empirical distribution of feature i ’s values as p_i^{norm} . To derive the *normal value range* $\mathcal{R}_i^{\text{norm}} \subseteq \mathcal{U}_i$, we define it as the interval containing

$1 - \alpha$ of the cumulative probability mass of the distribution $\mathcal{R}_i^{\text{norm}} = [\mu_i^{\text{norm}} - z_{\alpha/2}\sigma_i^{\text{norm}}, \mu_i^{\text{norm}} + z_{\alpha/2}\sigma_i^{\text{norm}}]$, where μ_i^{norm} and σ_i^{norm} denote the mean and standard deviation of feature i over CQRIs, and $z_{\alpha/2}$ is the standard normal quantile for confidence level $1 - \alpha$. This technique corresponds to the standard confidence interval construction for unimodal distributions and is widely adopted in statistical literature [15].

Similarly, for each adversarial technique $\mathcal{T}_j$, we model the distribution of feature i across QR codes identified to exhibit this technique, denoted $p_{i,j}^{\text{adv}}$, and derive its adversarial value range using the same high-confidence interval estimation: $\mathcal{R}_{i,j}^{\text{adv}} = [\mu_{i,j}^{\text{adv}} - z_{\alpha/2}\sigma_{i,j}^{\text{adv}}, \mu_{i,j}^{\text{adv}} + z_{\alpha/2}\sigma_{i,j}^{\text{adv}}]$. The *anti-adversarial range* excludes values statistically associated with adversarial perturbations. We define it as the set complement of the adversarial range over the universe $\mathcal{R}_{i,j}^{\text{anti}} = \mathcal{U}_i \setminus \mathcal{R}_{i,j}^{\text{adv}}$.

To determine whether feature i is significantly affected by adversarial technique $\mathcal{T}_j$, we compute their correlation $C_{i,\mathcal{T}_j}$ as *Bhattacharyya distance* between the normal and adversarial feature distributions $C_{i,\mathcal{T}_j} = \text{BD}(p_i^{\text{norm}}, p_{i,j}^{\text{adv}}) = \frac{1}{4} \ln \left(\frac{1}{4} \left(\frac{\sigma_i^{\text{norm}}}{\sigma_{i,j}^{\text{adv}}} + \frac{\sigma_{i,j}^{\text{adv}}}{\sigma_i^{\text{norm}}} + 2 \right) \right) + \frac{1}{4} \left(\frac{(\mu_i^{\text{norm}} - \mu_{i,j}^{\text{adv}})^2}{\sigma_i^{\text{norm}} + \sigma_{i,j}^{\text{adv}}} \right)$, assuming both distributions are approximately Gaussian. The Bhattacharyya distance is widely used for quantifying the separability of two probability distributions [49]. A large Bhattacharyya distance (e.g., exceeding a threshold τ) indicates that feature i is strongly influenced by $\mathcal{T}_j$.

- **Adversarial Technique Attribution:** Given the modeled ranges, we define binary identification functions for each technique $\mathcal{T}_j$. For an input image x with feature vector $\mathbf{v}(x)$, we say that feature i indicates technique $\mathcal{T}_j$ if $M_{i,j}(\mathbf{v}(x)) = \mathbb{I}[v_i(x) \in \mathcal{R}_{i,j}^{\text{adv}}]$, where $\mathbb{I}[\cdot]$ is the indicator function. Then, x is identified as exhibiting technique $\mathcal{T}_j$ if any of the features correlated with $\mathcal{T}_j$ falls into its adversarial range $\mathcal{T}_j(x) = \bigvee_{C(i,\mathcal{T}_j) > \tau} M_{i,j}(\mathbf{v}(x))$. To obtain the complete set of adversarial techniques associated with an image x , we evaluate each technique-specific identification function $\mathcal{T}_j(x)$ across all defined techniques $\mathcal{T}_1, \dots, \mathcal{T}_K$. Specifically, we define $\mathcal{T}(x) = \{\mathcal{T}_j \mid \mathcal{T}_j(x) = 1, j = 1, 2, \dots, K\}$. Here, $\mathcal{T}(x) \subseteq \{\mathcal{T}_1, \dots, \mathcal{T}_K\}$ denotes the set of adversarial techniques identified in image x , based on whether any correlated feature falls into the corresponding adversarial range.

This formulation supports multi-label detection, reflecting the fact that a single AQRI often exhibits multiple adversarial techniques simultaneously. The resulting set $\mathcal{T}(x)$ provides the foundation for subsequent targeted AQRI restoration.

4.4 Robust Decoder: Restoration and Reliable Decoding

Technique-Specific Feature Restoration. To restore an AQRI x , we aim to adjust its feature values so that none fall within the adversarial ranges of the identified techniques, including two phases.

- **Shared Feature Adjustment.** For each feature i that is

correlated with multiple detected techniques $\text{Tech}(i) = \{\mathcal{T}_j \mid C(i, \mathcal{T}_j) > \tau\}$, we compute the restoration target range $\mathcal{R}_i^{\text{restore}} = \mathcal{R}_i^{\text{norm}} \cap \bigcap_{\mathcal{T}_j \in \text{Tech}(i)} \mathcal{R}_{i,j}^{\text{anti}}$. If the intersection is non-empty, Adato adjusts the image using tailored image processing operations (e.g., histogram equalization, deblurring) to move $v_i(x)$ into $\mathcal{R}_i^{\text{restore}}$. Otherwise, the system raises an alert indicating that the feature distribution of the image exhibits abnormal characteristics inconsistent with any known anti-adversarial range, suggesting a potentially novel or heavily obfuscated adversarial technique.

- **Single-Technique Feature Adjustment.** For features correlated with only one technique, we adjust $v_i(x)$ to fall within $\mathcal{R}_i^{\text{norm}} \cap \mathcal{R}_{i,j}^{\text{anti}}$ for the corresponding $\mathcal{T}_j$. After both phases, the resulting image x' satisfies $\forall j, \forall i$ correlated with $\mathcal{T}_j, v_i(x') \notin \mathcal{R}_{i,j}^{\text{adv}}$, indicating successful mitigation of adversarial characteristics.

Reliable Decoding. Following the restoration processing of AQRI, we applied the robust decoding tool of WeChat [93] to parse all restored AQRI. Since Adato had already analyzed, identified, and mitigated known adversarial techniques to improve image scannability, this step aimed to verify the effectiveness of the restoration process through direct decoding. In cases where decoding still failed, we attributed the failure either to the presence of unknown adversarial techniques beyond our current taxonomy or to incomplete restoration. For such instances, we adopted a brute-force strategy that iteratively adjusted image parameters until a successful scan was achieved. The procedure of Adato for technique-specific restoration and decoding is detailed in Appendix A.

4.5 Evaluation of Adato

Implementation. Adato is deployed on a Linux server with 56 CPU cores, 377GB RAM, and 4 V100 GPUs. For QR code detector, we built a CNN YOLOv8 model using TensorFlow [23]. Next, for the AQRI identifier, we strictly followed official developer documentation for each tool, implemented the code to invoke them, and automatically obtained processing results from existing tools. We applied DBSCAN [27] for unsupervised clustering, and set the key hyperparameter ϵ as 31, by setting k as the number of features and $\epsilon = 2k - 1$, based on our experiment. To derive the feature distribution in CQRIs, we set $\alpha = 0.05$, corresponding to a 95% confidence interval. To determine whether a feature is significantly affected by an adversarial technique, we set the threshold value of Bhattacharyya distance as $\tau = 2.6$. Finally, in the restoration phase, we used threshold adjustment to restore pixels and decode them via WeChat’s QR code processing tool.

Datasets. To train, validate, and benchmark Adato, we prepare a comprehensive dataset suite spanning both controlled and real-world settings. This suite consists of two main parts: (1) a manually curated ground-truth dataset, and (2) a carefully annotated subset of real-world data for robust evaluation. Each dataset plays a distinct role in supporting various stages of

Table 5: Adato’s detection performance and restoration improvement for QR code detection tools on RealEval-QRC.

Tool	Original Recall* (%)	After-restored Recall* (%)	Improved by
Adato	97.8	–	–
Google Vision AI [36]	13.2	77.8	5.89×
Tencent Cloud [95]	15.1	80.6	5.33×
Baidu AI Cloud [8]	11.5	71.5	6.21×
Aliyun [4]	8.6	73.2	8.51×
Partner’s detector	62.1	90.3	1.45×

*: Recall means the success rate of AQRI detection.

system development and assessment.

- *Groundtruth Dataset*: To support Adato’s initial QR code verification, we constructed a labeled ground-truth dataset with three image categories: AQRI, CQRI, and non-QR code images. Specifically, 3,000 AQRI were collected from real-world user complaints (Section 3). In parallel, we randomly sampled 2,000 CQRI from a publicly available dataset [20], manually verifying that they followed standard QR code encoding practices and exhibited no signs of adversarial manipulation. To provide a contrasting set of non-QR content, we randomly selected 2,000 non-QR images from the raw collected dataset in Section 4.1. All images were independently labeled and cross-validated by two researchers. The resulting 7,000-image dataset was split into training, validation, and test sets using an 8:1:1 ratio and was used for training and evaluating the QR Code Detector.

- *Real-World AQRI Evaluation Set* (RealEval-QRC). To assess Adato’s detection and restoration capabilities in real-world scenarios, we constructed a manually labeled evaluation dataset from our large-scale image collection. Specifically, we randomly sampled 1,000 images from raw collected dataset in Section 4.1. We iteratively annotated sampled images until we obtained a target composition of 500 AQRI, 250 CQRI, and 250 non-QR code images, forming a manually curated dataset denoted as (RealEval-QRC). While this dataset adopts a balanced class distribution to enable systematic evaluation, it is drawn entirely from in-the-wild sources and thus retains the natural variability and noise of real-world data.

Evaluation of Adato. We evaluate Adato using RealEval-QRC and *Groundtruth Dataset*, first presenting step-wise results, then overall performance.

- *QR Code Detector*. On *Groundtruth Dataset*, it achieves 96.0% precision on the test set and 95.5% on the validation set. We further evaluated its performance on RealEval-QRC, where it demonstrates 98.6% precision, 97.8% recall, and an overall accuracy of 97.40% for AQRI detection. Furthermore, to benchmark its effectiveness, we compared Adato with several QR Code detectors. As reported in Table 5, Adato achieves an overall detection recall of 97.8%, significantly

Table 6: Robustness of Adato under N-M Out-of-Distribution Settings.

Setting	Training Data (%)	Precision (%)	Recall (%)
N-1	95.7	97.2	91.4
N-2	91.2	93.9	86.5
N-3	87.0	89.2	80.3
N-4	83.0	82.6	74.1
N-5	78.9	73.8	68.2
N-6	75.0	< 65.0	< 65.0

outperforming other models in detecting AQRI. Notably, Google Vision AI achieved only 13.2% recall on AQRI, highlighting Adato’s superiority in this challenging task. Expanding our analysis to out-of-distribution scenarios, we test Adato’s robustness in detecting AQRI with new forms of visual perturbations under N-M settings (M=1,...,6). By excluding M techniques from training (full results in Table 6), we observe that under the N-1 setting (95.7% of the training data), Adato maintains high performance, achieving 97.2% precision and 91.4% recall. A sharp drop (both metrics < 65%) occurs only in the N-6 setting (75.0% data). This confirms Adato’s strong generalization to out-of-distribution samples.

- *AQRI Identifier*. We evaluate the performance of the Adversarial Signal Detection component within Adato’s AQRI Identifier on RealEval-QRC. Adato achieves 93.3% precision, 91.2% recall, and 89.7% accuracy in identifying the presence of adversarial manipulations in QR code images. Furthermore, to evaluate Adato’s capability in identifying specific adversarial manipulations, we conduct a fine-grained performance breakdown across all 13 individual techniques. As detailed in Appendix C, Adato achieves consistently high precision, recall, and accuracy across diverse techniques, demonstrating its robustness in isolating distinct adversarial attacks.

- *Robust Decoder*. Adato achieves a reliable decoding success rate of 99.87% on RealEval-QRC after the restoration process. Upon manual inspection, the decoding failures came from two primary sources: non-functional QR codes (e.g., icon-based or SVG types) that lacked both decodability and valid information, and images with severe damage that were thus inherently undecodable.

- *Effectiveness as a pre-adaptor*. Based on the calculated adversarial feature range, we utilize the final two stages of Adato as a pre-adaptor to improve the effectiveness of various third-party QR code detectors. As shown in Table 5, detectors such as Google Vision AI, Baidu AI Cloud, and Tencent Cloud experienced recall improvements of 5 to 8.5 times over their baselines. Furthermore, by deploying the pre-adaptor within our partner’s proprietary detector, we achieved a state-of-the-art recall of 90.3%. These Results demonstrate the generality and effectiveness of Adato’s visual restoration design.

We further explored the efficiency of Adato and pre-adaptor (see Table 7): for Adato, it takes 52.66 minutes for 10,000 images, demonstrating efficiency in handling massive image

Table 7: Processing Efficiency of Adato.

Stage	Runtime (min)	Images per minute
Data Crawling	5.21	1919
QR Code Detector	7.67	1304
AQRI Identifier	26.28	381
Robust Decoder	13.50	741
Overall	52.66	190

data from social media platforms. As the pre-adaptor involves only the final two stages, with no need for range calculation and decoding, its runtime is reduced to 19.68 minutes.

5 Real-World Impact of AQRI

In this section, based on Adato’s detection results on the large-scale real-world dataset, we conducted a systematic measurement study on AQRI to reveal the current status of real-world usage and abuse.

5.1 Overall Dissemination of AQRI

By legally crawling image-containing posts published on 5 social media platforms, We obtained a total of 40,147,738 images contained in 24,311,446 posts. First, we identified 1,585,706 QR code-bearing images, with an average of 1.06 per post. Among these, 68,467 (4.32%) images were identified as AQRI, across 64,283 posts. After the restoration and decoding of AQRI, we successfully parsed 68,379 valid embedded information, including 68,331 URLs and 48 text messages. Additionally, 4,018 accounts were found to post AQRI. Of these accounts, 51.29% released at least five posts, and the maximum post count for a single account was 180.

Finding 1: Real-world results indicate that AQRI are already widely used as dissemination vectors on social platforms, with active posting activities.

5.2 Adversarial Techniques in AQRI: Image Processing & Generative AI

Understanding the adversarial techniques employed in AQRI is essential for uncovering how they exploit vulnerabilities in current moderation pipelines and for explaining the empirical disparities observed in detection and decoding. Based on the clustering analysis in Section 4.3, we identify 13 distinct categories of adversarial techniques. Each category is named after its closest image processing operation, forming a structured AQRI taxonomy (Figure 3) that captures how AQRI disrupt QR code processing across four key dimensions: visual concealment, contrast degradation, structural distortion,

Table 8: Top 10 adversarial technique combinations.

Operation Category	Count
Color Overlay + Edge Erosion + Noise Addition + Inversion	11,736
Blurring + Haze Overlay + Geometric Deformation + Overexposure + Contrast Blending	7,694
Blurring + Haze Overlay + Overexposure + Contrast Blending	4,241
Black Overlay + Color Overlay + Edge Erosion + Noise Addition + Inversion	3,354
Color Overlay + Edge Erosion + Noise Addition + Inversion + Contrast Blending	2,819
Color Overlay + Noise Addition + Inversion	2,124
Blurring + Color Overlay + Edge Erosion + Inversion	1,591
Blurring + Haze Overlay + Overexposure + Contrast Blending + Inversion	1,504
Color Overlay + Edge Erosion + Inversion	1,317
Blurring + Color Overlay + Inversion	1,146

and sharpness reduction. Further details on each technique are provided in Appendix D.

5.2.1 Image Processing for AQRI

Based on the identification of adversarial techniques, we analyzed the real-world usage, shown in Figure 4. First, we find that the most prevalent adversarial techniques in AQRI can be implemented via basic image processing operations, like Inversion (55.44%) and Color Overlay (48.65%). Unlike traditional adversarial attacks that rely on complex optimization [67, 68, 84, 109], AQRI also use QR-specific obfuscation strategies, such as Geometric Deformation (20.43%) and Density Compression (1.25%). This preference likely reflects the need to maintain decodability.

Furthermore, we found that using multiple techniques is more common in AQRI. Specifically, as shown in Figure 5, only 13.60% used a single adversarial technique. The maximum number of combined techniques reached 8 (covering 411 AQRI). Notably, 4 techniques formed the most common combination, covering 22,457 AQRI (32.80%). We further analyzed the most common combination strategies, as shown in Table 8. We find common combinations typically employ techniques across multiple key dimensions, as illustrated in Figure 6. This combination strategy, we argue, significantly increases adversarial effect complexity, thereby improving bypass success rates.

Finding 3: To maintain decodability, adversarial techniques in AQRI are tailored to QR-specific scenarios rather than general complex adversarial techniques, most implemented via basic image processing operations. Though individual techniques are basic, combining multiple ones yields complex adversarial effects, ensuring a high evasion success rate at minimal cost.

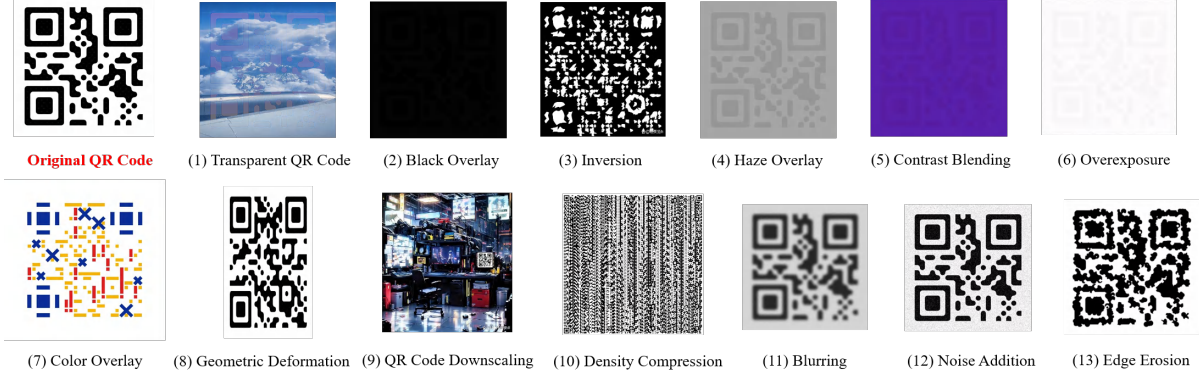


Figure 3: Examples of each adversarial technique in AQRI.

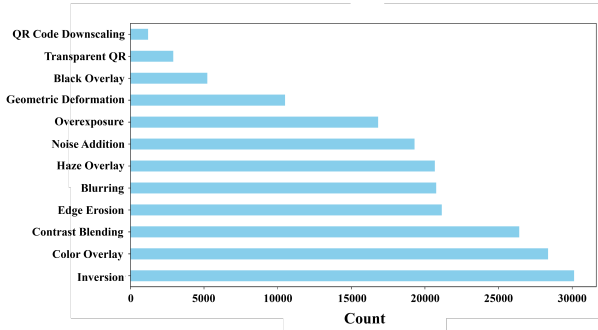


Figure 4: Adversarial techniques used in AQRI.

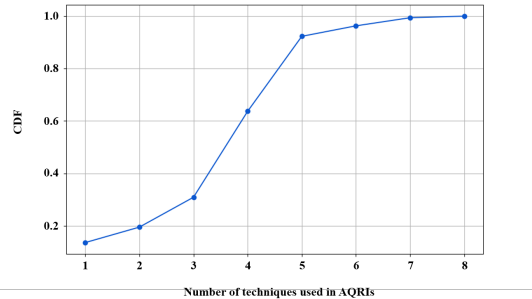


Figure 5: CDF of Adversarial Technique Usage Count in AQRI.

5.2.2 Generative AI in AQRI

The creation of AQRI is no longer limited to manual adversarial transformations, as adversarial image generation has become increasingly scalable and sophisticated with the rise of large-scale generative models [58]. Our analysis of in-the-wild AQRI from Adato reveals that adversaries have begun incorporating AI-generated imagery (AIG) to construct what we term *Generative AQRI* (as shown in Figure 6), which

often embed multiple adversarial effects within a single image, enabling more complex and evasive attacks. To identify such content and assess its impact, we use Hive’s AI Detector [42], which detects AI-generated images and estimates their likely source models, offering insights into the underlying generative techniques. This tool demonstrated 99% accuracy on a validation set of 100 Generative and 100 non-AI AQRI during our small-scale verification experiment.

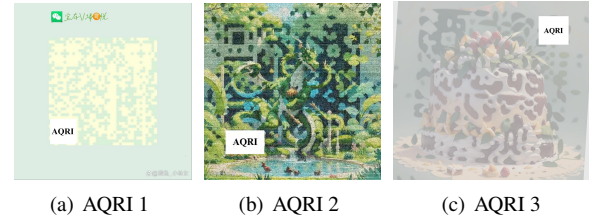


Figure 6: Examples of AQRI: (a) combining multiple adversarial techniques, (b) utilizing Generative AQRI, and (c) integrating adversarial techniques with Generative AQRI. Part of each image is obscured to prevent the dissemination of malicious content.

Prevalence of Generative AQRI. Among the 68,467 in-the-wild AQRI identified in Section 5.1, 5,377 (7.85%) were constructed using generative AI, with 98.89% attributed to Stable Diffusion [82], making it the predominant tool for generating AI-driven AQRI. The remaining cases involved models such as Wan [104] and Kling [52]. Notably, we find no evidence of advanced large vision models (LVMs) such as Sora [70], Nano Banana [37], or Gork [111] being employed in this dataset, indicating that adversaries currently rely on general-purpose generative models, with limited adoption of more specialized vision architectures. Over our six-month collection period, the monthly volume of Generative AQRI surged by 350% (from 282 cases in September 2024 to 1,269 cases in March 2025), indicating a rapid escalation in their

adoption.

Bypass Performance. Generative methods naturally blend multiple adversarial effects in AQRI, with common technique combinations (Table 8) frequently observed. For example, 4,836 (89.94%) Generative AQRI exhibit the most prevalent combination: Color Overlay, Edge Erosion, Noise Addition, and Inversion. To assess evasion, we randomly sampled 100 Generative and 100 non-AI AQRI and tested them against the detectors in Table 1. Generative AQRI achieved a significantly higher average bypass rate of 89.0%, compared to 69.0% for non-AI AQRI. Surprisingly, advanced detectors based on large vision models (LVMs), including Baidu AI Cloud [8] and Aliyun [4], performed worse on Generative AQRI, yielding bypass rates of 87.0% and 69.0%, versus 72.0% and 31.0% for non-AI counterparts. Manual inspection shows that Stable Diffusion often generates photorealistic backgrounds that distract model attention from the QR code, causing misclassification as artistic content. Generative AI thus produces more sophisticated AQRI, making QR codes harder to detect and moderation more difficult.

Abuse Potential: Accessibility, Restriction, and Decodability. To evaluate the feasibility of fabricating Generative AQRI, we examined publicly available generative services based on Stable Diffusion. We collected the 20 most-starred Stable Diffusion-based QR code generators on GitHub, 16 of which offered web interfaces for direct use. First, manual review of README files and service policies showed that none of the 20 repositories or 16 web interfaces imposed restrictions or content checks on the QR payload, allowing unrestricted use and potential abuse for generating policy-violating AQRI. Second, several generators, such as QRBTf [55], offered extensive prompts, visual customization options, and AI tutorials, significantly lowering the barrier to use. Additionally, generators like QRcode-antfu [6] include decodability verification to optimize readability, supporting the threat model by enabling reliable information delivery under evasion. Our decodability evaluation on 100 Generative and 100 non-AI AQRI across multiple decoders showed that Generative AQRI consistently achieved higher decoding success, with an average margin of 18%, likely due to the greater flexibility of generative methods in refining QR structures. Given their superior bypass performance, improved decodability, and ease of fabrication, Generative AQRI constitute a compelling upgrade for adversaries. While still a minority of all AQRI, the maturity of the technology and accessibility of supporting services make them an emerging vector for abuse.

Evaluating LVMs for Generating AQRI. Although we found no evidence of LVM use in our dataset, their ability to generate highly realistic images raises concerns about potential abuse for AQRI generation. To assess this, we selected 3 leading LVMs (Nano Banana, Sora, and Grok) and created 20 prompts based on the styles of 20 in-the-wild Generative AQRI. Each model generated 20 AQRI from a controlled CQRI, yielding 60 images in total. Manual inspection showed

that Nano Banana and Grok preserved QR structures, while 40% of Sora’s outputs failed and were discarded. Bypass evaluation achieved a 73.5% average success rate—higher than non-AI AQRI but lower than Stable Diffusion. Decodability remained a major limitation: only 8 of 60 images (13.3%) were readable. This illustrates what we call “Functional Blindness” in LVMs: they replicate the visual appearance of AQRI sufficiently to evade detection but fail to preserve the syntactic structure needed for decoding. Unlike Stable Diffusion, which refines existing CQRI, LVMs generate content from scratch and often ignore decoding constraints, making them currently ineffective for producing functional AQRI.

Finding 4: Generative AQRI, primarily created via Stable Diffusion, are highly evasive, reliably decodable, and easily accessible given unrestricted usage policies and low barriers to use. Their ability to blend multiple adversarial effects while preserving functionality poses a growing threat to existing moderation systems. In contrast, large vision models remain ineffective for functional AQRI generation due to poor decodability.

5.3 Information Behind AQRI

To uncover their adversarial intent, we decoded 68,467 AQRI via Adato and extracted valid content from 68,379 of them. Of these, 99.80% (68,331) embedded URLs, indicating AQRI are primarily used to redirect to promotional websites. Analyzing the remaining 48 text-encoded AQRI, we found that 15 of these entries were fraudulent, including spammer contacts and fraudulent promotional content.

URL Category. URLs (99.80%) are the primary information embedded in AQRI, so we analyzed them further. Deduplication yielded 2,136 distinct URLs, including 861 fully qualified domain names (FQDNs) and 639 second-level domain names (SLDs). Manual inspection of these domains and their webpage content identified four domain types associated with AQRI activities. An initial double-coding of 300 randomly selected URLs (14%) achieved 100% agreement, allowing the remainder to be confidently annotated by a single researcher. First, 501 short URLs (23.46%) involve 13,642 AQRI, mainly from 6 services (e.g., Twitter, Weibo, Google, ibit.com). Short links enable auto-redirection and are widely used to hide malicious links [121], increasing the moderation difficulty of AQRI and their embedded info. Of these, 407 URLs (10,281 AQRI) provide porn app downloads (covert channels), and 83 (2,011 AQRI) offer illegal LLM-based services (e.g., porn chatbots, drug sales). Second, 406 URLs (19.01%) belong to 5 major social media platforms (e.g., WeChat, Instagram), linking to 8,019 AQRI. These URLs typically redirect to platform-specific accounts or posts. In-depth analysis shows linked accounts engage in illegal services: fraud (e.g., fake investments, pig-butcher scams) and porn promotion. Three confirmed Instagram accounts

promote pornography/gambling sites, with 293k followers and 5,896 AQRI posts. Notably, 235 URLs (11.00%) from government/educational domains involve 6,667 AQRI. Adversaries exploit these sites’ vulnerabilities to compromise; they deploy AQRI to leverage authoritative credibility for covert promotion [87, 120]. Finally, personally built sites⁴ are the most common, accounting for 994 URLs (46.54%). 99.95% of these sites host malicious payloads spanning the seven categories detailed below.

URL Content Business. To further reveal AQRI-propagated content, we identified 8 scenarios from related URLs (by domain, post, and webpage content). Legitimate promotion is rare (2.67% of URLs, e.g., personal websites). The remaining 2,079 URLs (97.33%) link to 65,578 AQRI (95.78%) promoting malicious activities, covering 7 types: Pornography (65.12%), gambling (29.63%), account trading (2.45%), drug sales (1.14%), proxy services (0.76%), crypto sales (0.60%), and malware delivery (0.30%). Payload distributions exhibit clear cross-platform divergence. Pornographic content concentrates heavily on Baijiahao and Tieba, comprising 77.6% of the malicious AQRI detected on these platforms. In contrast, gambling and drug-related promotions prevail on Twitter, Instagram, and Reddit, accounting for 80.2%, 52.3%, and 69.8% of their respective AQRI volumes. Despite active malicious AQRI propagation, professional link detectors largely overlook them. For example, VirusTotal [41] flagged only 143 URLs as malicious. This stems from AQRI’s diverse URL types (e.g., social platforms, educational sites), inherent stealth, and low attention.

Finding 5: Analysis of embedded info shows AQRI is widely abused in illegal businesses. Online pornography is most common; notably, government/educational domains are misused to spread AQRI.

5.4 Information Dissemination Strategies

To unveil the strategy of AQRI dissemination, we analyze AQRI posts to identify traffic-driving and user-attraction techniques, and uncover methods to obfuscate illicit operations, optimize evasion, and circumvent content-layer censorship.

Strategy 1: Additional Adversarial Signal. Promoters employ multi-modal evasion techniques. First, we utilized Google Vision AI [36] to classify image content and identify specific threats (e.g., porn). Manual verification confirmed that attackers frequently mask illicit intent behind benign imagery, such as cartoons redirecting to adult content. Besides, adversaries also embed and obfuscate extra text in QR images. For instance, “微信扫描 (WeChat scan)” may be altered to “V-扫 (V-Scan)” by replacing “微” with “V” (see Figure 7). Finally, adversaries forge trust indicators to simulate authenticity. Some QRs embed counterfeit icons from trusted platforms

⁴We define personally built sites as domains independently registered and hosted by individual developers.

(e.g., fake WeChat QRs), misleading users into believing they are legitimate while bypassing actual WeChat services.

Strategy 2: Mimicking Normal Behavior. To evade moderation during dissemination, adversaries mimic legitimate user behavior. For example, 54.96% of posts contain only two images, imitating casual sharing to evade “marketing account” detection, and 69.34% are posted after 21:00, likely exploiting perceived weaker nighttime moderation. Second, adversaries exploit cross-platform differences in QR moderation to construct indirect propagation paths. Specifically, AQRI-based malicious content is posted on low-moderation platforms (e.g., Baidu Tieba) and users are guided to scan it using high-decoding platforms (e.g., WeChat), enabling redirection and dissemination of malicious resources. This gap turns weaker platforms into “entry intermediaries” in the attack chain (see Figure 8).



Figure 7: Examples of adversarial text usage in AQRI.

Strategy 3: Enhancing Redirection Challenge. When directing users to malicious resources, adversaries adopt multi-dimensional evasion strategies. First, most platforms inspect only initial QR redirects, leaving multi-level chains unchecked. Adversaries exploit this gap by creating long redirect paths. We observed heavily tampered QRs leading to intermediate pages with slightly modified QRs, which further redirected to a WeChat account that automatically forwarded users to illegal gambling sites. Access restrictions are also employed: 136 URLs used cloaking to hide malicious content from moderation crawlers. Besides that, they exploit or forge

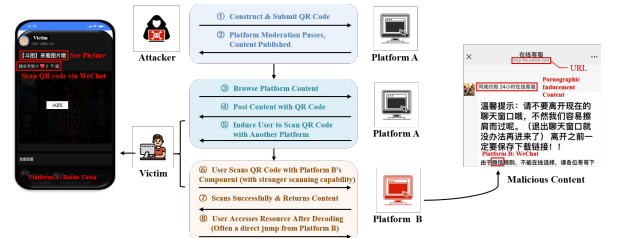


Figure 8: Workflow of Indirect Propagation Path

legitimate info to mislead detectors. They illegally exploited

legitimate domains (e.g., government/educational or server-less platforms [62]) to host AQRI; e.g., 3,529 AQRI were found, uploaded to such sites with porn content. Additionally, promoters hide malicious links behind normal images via short links, extending redirection chains to complicate auditing.

Finding 6: AQRI dissemination activities show diverse adversarial promotion strategies to further increasing moderation difficulty.

6 Discussion

Method Limitations. Despite efforts in method design and implementation, several limitations remain. First, Adato identifies QR codes using finder patterns according to specifications. This can produce false positives when images contain similar patterns, such as grid-like windows. Manual analysis of 2,000 randomly sampled images from the QR code detector on the In-the-Wild Dataset found only 11 false positives, indicating a negligible impact due to the very low false positive rate. Moreover, these misidentified images lack decodability and are filtered out in step 3 (Robust Decoder), leaving results unaffected. Second, we cluster AQRI using 16-dimensional features to identify adversarial techniques. Although this may introduce some errors, a precision of 93.3% ensures their impact on the analysis is minimal.

Responsibly Abuse Threat Disclosure. Based on our measurements, we disclosed AQRI-related risks to affected social media platforms by reporting detected AQRI and post links, and summarizing potential moderation gaps revealed by differences in adversarial techniques. Several vendors responded positively, and we shared our enhanced tool, Adato, as an adaptor to support their moderation mechanisms.

Lessons Learned and Mitigation. Our work is the first to systematically reveal real-world AQRI, which are severely abused in illegal promotions-showing current moderation mechanisms fail to address emerging AQRI risks. Based on our findings, we propose two mitigation recommendations.

- *Enhance moderation capabilities against AQRI.* Active testing and risk disclosure show that current moderation mechanisms remain largely unaware of AQRI risks. Our proposed tool, Adato, can serve as a pre-enhancer for existing QR code tools, improving moderation by detecting and correcting adversarial perturbations. We will open-source Adato along with the AQRI ground-truth dataset for research use.
- *Enhance user prompts.* QR code moderation must consider both the image and its embedded resources. Some platforms (e.g., Zhihu [122]) lack security prompts for QR-induced redirections. Unlike direct links, QR codes prevent users from previewing embedded content, allowing adversaries to use AQRI to bypass moderation and deliver malicious content. Therefore, platforms should enhance QR code detection and provide effective user prompts before scanning.

7 Related Work

QR Code Detection. QR codes are increasingly abused in malicious activities, e.g., malicious promotion [117] and covert attacks [7, 39, 59, 107, 114]. For effective defense, existing work focuses on enhancing QR code detection. Liao et al. [60] proposed a training-free QR code generator based on Stable Diffusion. Vinay et al. [25] designed a fast detection scheme combining progressive discrimination with MobileNet to boost practical accuracy. *However, current techniques severely lack AQRI detection capabilities, enabling AQRI to bypass existing mechanisms effectively.*

Image Content Moderation. Image content moderation is key to securing social media security, with ML-based image processing models as primary technical means. Platzer et al. [77] proposed ML-based skin detection for porn images, while Yuan et al. [117] introduced R-CNN-based methods for illicit promotion image detection. For videos, Wehrmann et al. [106] classified adult content via CNN-LSTM fusion, and Perez et al. [75] improved accuracy by combining static and motion information. *However, these methods neglect the impact of adversarial techniques applied to QR Codes.*

Adversarial Attack and Defense. Adversarial attacks add subtle perturbations to images, misleading AI models into misclassification while preserving human recognition. Their generation methods evolved from gradient-based approaches [22, 34, 47, 54, 64, 68, 84, 115], to boundary exploration methods for black-box scenarios [13, 16, 17, 19, 43, 44, 61, 67, 73, 74]. Wu et al. [109] further explored ways to enhance adversarial example transferability. Given the attack capability of adversarial examples on AI models, their defense is a research focus. Qin et al. [78] used class-conditional reconstruction for detection, noting CapsNet outperforms CNNs in perceptual alignment to aid this. Li et al. [57] leveraged contextual inconsistency of adversarial patterns in images for external detection. Yin et al. [116] introduced generative adversarial training to learn a detector, enhancing robustness against adaptive attacks via asymmetric training. *However, existing research on adversarial examples targets general scenarios. The unique structure of QR codes makes existing generation and defense methods hard to apply directly to AQRI.*

8 Conclusion

Adversarial QR Code Images (AQRI) emerge to bypass current social media platforms' QR code moderations by adding perturbations to conventional QR code images (CQRIs). Our work is the first to systematically measure this emerging threat. First, based on ground-truth dataset analysis, we designed a finder-pattern-based enhanced AQRI identification method, Adato, followed by restoring added adversarial techniques to ensure successful decoding. Using Adato on 40,147,738 images collected from 5 social media platforms, we detected 68,467 AQRI. We conducted the first real-world measure-

ment analysis on these AQRI, covering their adversarial techniques, embedded promotional information, campaign strategies, and promotional gains. Combined with responsible disclosure, we propose several mitigation recommendations, and will open-source our identification and restoration tool, Adato, to help enhance AQRI moderation capabilities.

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Ethical Considerations

First of all, through a formal collaboration with our partner, all research activities were conducted under the supervision of the partner’s legal department, partially compensating for the absence of an Institutional Review Board (IRB) at our own institution. We follow the ethical guidelines in the Belmont Report [29] and the Menlo Report [51], and proactively address three ethical considerations relevant to this study: (1) the use of user complaint data in Section 3; (2) active posting experiments in Section 3; (3) post and image data collection from five social media platforms in Section 4 and (4) researcher protection during the manual analysis of URLs behind AQRI in Section 5.

First, to collect real-world AQRI as our ground-truth dataset, we leverage large-scale user complaint data with the assistance of our partner. Although the complaint interface includes an explicit privacy protection statement, such data may still contain personally identifiable information (PII). To mitigate these ethical concerns, our partner performs manual review and anonymization (i.e., salted hashing) of all sensitive information before our data access. As a result, researchers never interact with any user PII and only analyze images and image-related post content from reported complaints.

Second, to accurately evaluate AQRI’s impact on moderation, we implemented active posting experiments in Section 3. However, on platforms that restrict QR codes, this could potentially conflict with their Terms of Service (ToS). Following legal advice, we conducted a stakeholder-aware risk assessment that explicitly accounted for potential harms to both platforms and users, and we therefore adopted the following strict mitigation measures: 1) we deleted all posts within one minute of submission, thereby testing only immediate moderation feedback; 2) each post included a disclaimer stating the experimental purpose and a contact email, and we committed to immediately terminating the experiment upon any user complaint; 3) if a post remained unbanned after one minute, we reported it before deletion to alert the platform; 4) we constructed the QR codes to mimic real AQRI exclusively at

the image level (i.e., visual appearance and distortion effects), ensuring all decoded payloads were entirely benign plain text devoid of any malicious URLs or harmful content. Crucially, as a simulated evaluation, these experiments exposed real-world vulnerabilities and can support the development of defenses against emerging risks; in this sense, the anticipated benefits plausibly outweighed the potential harms. We received no negative user feedback. Given the one-minute exposure window, the maximum view count was 6, with 87.9% of successful posts had only a single view (by the researcher), which further minimized potential impact.

Third, to measure the real-world prevalence and impact of AQRI at scale, we analyzed public posts containing images (including QR images) from five major social media platforms. For platforms with official data APIs (Reddit, Instagram, and Twitter), we legally purchased API access with the assistance of our partner and strictly complied with the relevant platform policies. For platforms without public APIs (Baidu Tieba and Baijiahao), we developed custom crawlers that adhere to robots.txt and platform-specific crawling policies. We further enforced rate limiting (e.g., minimum request intervals) to reduce server load and avoid violations of the platforms’ ToS, consistent with prior work [119]. Moreover, during data collection, we did not collect any sensitive user account information (e.g., usernames). Although public posts rarely contain explicit personally identifiable information (PII), we further mitigated privacy risks by anonymizing any potential PII that appeared in AQRI-related posts. We used all data solely for technical analysis and did not involve human-subject interaction. Our partner securely stored the data on physically isolated servers, with access restricted to authorized administrators. We have already disclosed all detected AQRI and posts to related social media platforms to help them timely identify and defend against the emerging AQRI risk, and we will permanently delete all raw data upon project completion. Overall, our partner’s legal team supervised all data collection, processing, and storage procedures to minimize potential ethical risks.

Finally, to protect researchers when accessing URLs behind AQRI, we implemented strict technical and psychological safeguards: 1) all accesses were conducted in an isolated virtual machine using a sandboxed browser (no persistent storage, script execution disabled) to prevent exposure to potential malware; 2) all researchers received specialized training prior to access. Upon encountering malicious content, we logged only categorical tags without storing any page resources. Recognizing the risk of psychological distress, researchers conducted the review at a self-managed pace with regular well-being check-ins and could pause the experiment at any time or access institutional counseling services if they experienced discomfort.

Responsible Disclosure. Beyond the above measures, we proactively disclosed all our experimental findings, including results from active probing experiments and large-scale mea-

surements, via their reporting channels and Security Response Centers (SRCs) of the five analyzed social media platforms. This disclosure aimed to raise awareness of AQRI-related risks and to provide platforms with concrete post-level evidence to facilitate timely mitigation. To date, we have received positive responses from two platforms. Furthermore, we reported the detection flaws to the evaluated vision API providers, documenting our contact dates and their responses.

Open Science

Our research adheres to open science principles to foster reproducibility and collaboration.

System components. We open-source the vast majority of code in Adato, except for the Data Collector which contains sensitive API credentials. All other components are published on <https://doi.org/10.5281/zenodo.20322613>. The Feature Restoration part can serve as an adapter for the preprocessing stage of existing recognition and scanning tools, significantly improving the performance of QR code detectors against AQRI. To ensure third-party users can utilize our work, we will launch a webpage to demonstrate our results. This webpage will feature an online interface for users to submit and detect AQRI, along with our contact information.

Ground-truth dataset. To help the community understand and defend against AQRI, we release the AQRI Ground-truth Dataset to enable threat analysis and improve detector training. However, due to the malicious nature of the pictures, the dataset is hosted on <https://doi.org/10.5281/zenodo.20325847> under Restricted Access.

Vulnerability disclosure. We are responsibly disclosing our findings to affected platforms and assisting them in deploying our enhancer to comprehensively mitigate this security risk. Therefore, the detection results are withheld because our vulnerability disclosure and platform remediation are still ongoing. Interested researchers may request the data via the contact information that will be provided on our webpage.

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A Adato Algorithm for Technique-Specific Restoration and Decoding

The procedure for adversarial technique attribution, AQRI restoration, and reliable decoding is detailed in Algorithm 1.

B Visual Metrics

Table 9 presents the interpretable visual metrics used in our research.

Table 9: Image Quality Assessment Parameters

No.	Metrics	Description
1	Aspect Ratio Deviation	Measures the deviation from standard aspect ratios (e.g., 1:1, 4:3, 16:9)
2	Area Proportion	Ratio of foreground area to total image area
3	Edge Complexity	Quantifies the complexity of edges in the image
4	Alpha Channel Ratio	Proportion of transparent pixels (if applicable)
5	V-Channel Peak Value (HSV)	Maximum value in the Value channel of HSV color space
6	LAB Color Difference Threshold	Threshold for perceptible color differences in LAB space
7	Mean HSV Saturation	Average saturation value in HSV color space
8	Number of Unique Colors	Count of distinct color values in the image
9	Laplacian Variance	Variance of Laplacian filtered image (sharpness measure)
10	Mean of Block-wise Standard Deviations	Average of standard deviations calculated in image blocks
11	Retinex Intensity	Intensity measure from Retinex algorithm
12	Brightness-to-Contrast Ratio	Ratio between overall brightness and contrast
13	Finder Pattern Region Proportion	Area proportion occupied by finder patterns (for QR/barcodes)

Algorithm 1 Adato: Adversarial Technique Attribution, Restoration, and Decoding

Input: Image x , Feature extractor $v(\cdot)$, Adversarial ranges $\mathcal{R}_{i,j}^{\text{adv}}$, Anti-adversarial ranges $\mathcal{R}_{i,j}^{\text{anti}}$, Normal ranges $\mathcal{R}_i^{\text{norm}}$, Correlation matrix $C(i, \mathcal{T}_j)$, Threshold τ

Output: Restored image x' , Decoded content info

```

1: // Step 1: Feature Extraction and Technique Attribution
2:  $v(x) \leftarrow v(x)$ 
3:  $\mathcal{T}(x) \leftarrow \emptyset$ 
4: for each technique  $\mathcal{T}_j$  do
5:   if  $\exists i$  such that  $C(i, \mathcal{T}_j) > \tau$  and  $v_i(x) \in \mathcal{R}_{i,j}^{\text{adv}}$  then
6:      $\mathcal{T}(x) \leftarrow \mathcal{T}(x) \cup \{\mathcal{T}_j\}$ 
7:   end if
8: end for
9: // Step 2: Feature-Based Restoration
10: for each feature  $i$  correlated with any  $\mathcal{T}_j \in \mathcal{T}(x)$  do
11:    $\text{Tech}(i) \leftarrow \{\mathcal{T}_j \mid C(i, \mathcal{T}_j) > \tau\}$ 
12:   if  $|\text{Tech}(i)| > 1$  then ▷ Shared Feature Adjustment
13:      $\mathcal{R}_i^{\text{restore}} \leftarrow \mathcal{R}_i^{\text{norm}} \cap \bigcap_{\mathcal{T}_j \in \text{Tech}(i)} \mathcal{R}_{i,j}^{\text{anti}}$ 
14:   else ▷ Single-Technique Feature Adjustment
15:     Let  $\mathcal{T}_j$  be the only element in  $\text{Tech}(i)$ 
16:      $\mathcal{R}_i^{\text{restore}} \leftarrow \mathcal{R}_i^{\text{norm}} \cap \mathcal{R}_{i,j}^{\text{anti}}$ 
17:   end if
18:   if  $\mathcal{R}_i^{\text{restore}} \neq \emptyset$  then
19:     Apply image transformation to adjust  $v_i(x) \rightarrow \mathcal{R}_i^{\text{restore}}$ 
20:   else
21:     Raise anomaly alert: feature  $i$  has no valid restoration range
22:   end if
23: end for
24:  $x' \leftarrow$  updated image after all adjustments
25: // Step 3: Reliable Decoding
26:  $\text{info} \leftarrow \text{Decode}(x')$  using robust decoder
27: if  $\text{info} = \emptyset$  then
28:   Apply brute-force decoding strategy.
29: end if
30: return  $x'$ ,  $\text{info}$ 

```

C Performance Breakdown by Technique

Table 10 provides a fine-grained performance breakdown of Adato across all 13 individual adversarial techniques.

D Adversarial Techniques in AQRI

Adversarial techniques disrupt QR codes along four key dimensions: *visual concealment* (Transparent QR Code, Black Overlay); *contrast degradation* (Inversion, Haze Overlay, Contrast Blending, Overexposure, Color Overlay); *structural distortion* (Geometric Deformation, QR Code Downscaling, Density Compression); and *sharpness reduction* (Blurring, Noise Addition, Edge Erosion). We elaborate on each technique below.

- 1) **Transparent QR Code:** Makes the QR code visually invisible by setting the pixel values within the QR code region to be transparent, thereby bypassing detections that solely rely on visible features.
- 2) **Black Overlay:** Overlays the QR code with a uniformly black region on the image, resulting in excessive darkness

Table 10: Fine-Grained Performance Breakdown Across 13 Adversarial Techniques

Technique*	Precision (%)	Recall (%)	Accuracy (%)
(1)	95.2	93.1	92.5
(2)	91.4	89.5	88.6
(3)	96.1	94.2	93.4
(4)	88.5	86.4	85.7
(5)	94.7	92.8	92.1
(6)	93.2	90.5	90.1
(7)	92.8	91.7	90.5
(8)	97.5	95.6	95.4
(9)	87.6	85.1	84.6
(10)	95.8	94.0	93.5
(11)	90.5	88.2	87.8
(12)	96.3	93.8	93.0
(13)	93.3	90.7	88.9
Overall	93.3	91.2	89.7

* : See Appendix D for technique details.

that renders the QR code structure visually indiscernible.

- 3) **Inversion:** Reverses the pixel distribution by swapping the foreground and background colors, disrupting detections that rely on grayscale conversion or binarization.
- 4) **Haze Overlay:** Reduces the contrast of the original image by applying a haze-like overlay, simulating a foggy or washed-out appearance that weakens the visual distinction between QR code modules and the background.
- 5) **Contrast Blending:** Reduces the visual contrast between the QR code and its background by making their pixel values closely similar, leading to failures in grayscale conversion or binarization processes that rely on clear foreground-background separation.
- 6) **Overexposure:** Overlays a white layer onto the image, producing an overexposed effect that reduces visual contrast and obscures the structural features.
- 7) **Color Overlay:** Overlaying the QR code region with additional or altered hues that break the standard black-and-white structure and interfere with detection systems that rely on binarization or fixed color expectations.
- 8) **Geometric Deformation:** Alters the geometric structure of the QR code by modifying the image’s aspect ratio, resulting in a flattened or stretched appearance that can impair detection expecting a standard square format.
- 9) **QR Code Downscaling:** Reduces the visual prominence of the QR code by scaling it down within a larger image canvas, decreasing its relative size and making it more difficult to localize.
- 10) **Density Compression:** Increases the QR code’s encoding vision, resulting in higher structural complexity and a

denser arrangement of modules, which reduces the relative size and visual prominence of finder pattern regions.

- 11) **Blurring:** Applies a blur effect to the entire image or specifically to the QR code region, softening edges and reducing structural clarity.
- 12) **Noise Addition:** Introduces random pixel-level variations either across the entire image or specifically within the QR code region to degrade structural clarity.
- 13) **Edge Erosion:** Degrades the internal edges and boundaries of the QR code, distorting its geometric consistency and weakening the structural features.